# Problem Statement

Evaluate the limit from the image "dailyintegral-limit-limit-1x1-2026-07-25.jpg":

$$L = \lim_{n\to\infty} \int_0^{\ln(n)} \frac{2 - e^x - e^{-nx}}{e^x - 1}\, dx$$

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# Detailed Solution

To evaluate this limit, we decompose the integrand algebraically, apply a variable substitution to simplify the resulting fraction, and utilize the asymptotic properties of the harmonic numbers.

## Step 1: Algebraic Manipulation of the Integrand

We begin by rewriting the numerator of the integrand to separate the exponential terms:

$$\frac{2 - e^x - e^{-nx}}{e^x - 1} = \frac{(1 - e^x) + (1 - e^{-nx})}{e^x - 1} = -1 + \frac{1 - e^{-nx}}{e^x - 1}$$

Substituting this identity back into our definite integral allows us to split it into two simpler integrals:

$$\begin{aligned}\int_0^{\ln(n)} \frac{2 - e^x - e^{-nx}}{e^x - 1}\, dx &= \int_0^{\ln(n)} (-1)\, dx + \int_0^{\ln(n)} \frac{1 - e^{-nx}}{e^x - 1}\, dx \\ &= -\ln(n) + \int_0^{\ln(n)} \frac{1 - e^{-nx}}{e^x - 1}\, dx\end{aligned}$$

## Step 2: Substitution and Integration

Let $I_n$ denote the second integral:

$$I_n = \int_0^{\ln(n)} \frac{1 - e^{-nx}}{e^x - 1}\, dx$$

We apply the substitution $u = e^{-x}$, which implies $x = -\ln(u)$ and $dx = -\frac{1}{u}\, du$. The limits of integration transform as follows:

- When $x = 0$, $u = e^0 = 1$.
- When $x = \ln(n)$, $u = e^{-\ln(n)} = \frac{1}{n}$.

The denominator can be expressed in terms of $u$ as:

$$e^x - 1 = \frac{1}{u} - 1 = \frac{1 - u}{u}$$

Substituting these into $I_n$, we obtain:

$$I_n = \int_1^{1/n} \frac{1-u^n}{\frac{1-u}{u}} \left(-\frac{1}{u}\,du\right) = \int_{1/n}^1 \frac{1-u^n}{1-u}\,du$$

Using the standard finite geometric series identity $\frac{1-u^n}{1-u} = \sum_{k=0}^{n-1} u^k$, we can integrate term by term:

$$I_n = \int_{1/n}^1 \left(\sum_{k=0}^{n-1} u^k\right) du = \sum_{k=0}^{n-1} \int_{1/n}^1 u^k\,du$$

$$= \sum_{k=0}^{n-1} \left[\frac{u^{k+1}}{k+1}\right]_{1/n}^1 = \sum_{k=0}^{n-1} \frac{1}{k+1}\left(1 - \frac{1}{n^{k+1}}\right)$$

Shifting the index of summation by setting $m = k+1$, we get:

$$I_n = \sum_{m=1}^{n} \frac{1}{m} - \sum_{m=1}^{n} \frac{1}{mn^m}$$

## Step 3: Connection to Harmonic Numbers

Recall that the $n$-th harmonic number is defined as $H_n = \sum_{m=1}^{n} \frac{1}{m}$. Combining this with our result from Step 1, the full definite integral before taking the limit is:

$$\int_0^{\ln(n)} \frac{2 - e^x - e^{-nx}}{e^x - 1}\,dx = H_n - \ln(n) - \sum_{m=1}^{n} \frac{1}{mn^m}$$

## Step 4: Asymptotic Evaluation and Limit

Now, we evaluate the limit as $n \to \infty$:

$$L = \lim_{n\to\infty} \left(H_n - \ln(n) - \sum_{m=1}^{n} \frac{1}{mn^m}\right)$$

We analyze the convergence of the two components independently:

1. By definition, the asymptotic difference between the harmonic number $H_n$ and the natural logarithm $\ln(n)$ converges to the Euler-Mascheroni constant, $\gamma$:

$$\lim_{n\to\infty} (H_n - \ln(n)) = \gamma \approx 0.5772156649\ldots$$

2. For the remaining summation, we can bound it above by an infinite geometric series for all $n \geq 2$:

$$0 < \sum_{m=1}^{n} \frac{1}{mn^m} \leq \sum_{m=1}^{\infty} \frac{1}{n^m} = \frac{\frac{1}{n}}{1 - \frac{1}{n}} = \frac{1}{n-1}$$

By the Squeeze Theorem, since $\lim_{n\to\infty} \frac{1}{n-1} = 0$, we have:

$$\lim_{n\to\infty} \sum_{m=1}^{n} \frac{1}{mn^m} = 0$$

Combining these evaluations yields the final value of the limit:

$$L = \gamma - 0 = \gamma$$

**Final Answer:**

$$\gamma$$